higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(First Paper)
NQF LEVEL 3

20 FEBRUARY 2019

This marking guideline consists of 10 pages.

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QUESTION 1

1.1 1.1.1

$$\begin{aligned}&(5-2i)^2\\&=(5-2i)(5-2i)\\&=25-10i-10i+4i^2 \checkmark\\&=25-20i+4(-1) \checkmark\\&=25-20i-4\\&=21-20i \checkmark\end{aligned}$$

(3)

1.1.2

$$\begin{aligned}&(12+4i)-i^3(6-3i)\\&=12+4i-6i^3+3i^4 \checkmark\\&=12+4i-6(i^2)i+3(i^2)^2\\&=12+4i-6(-1)i+3(-1)^2 \checkmark\\&=12+4i+6i+3 \checkmark\\&=15+10i \checkmark\end{aligned}$$

(4)

1.1.3

$$\begin{aligned}&7i^3-2-3i-8i^2+\sqrt{16}\\&=7(i)^2i-2-3i-8(-1)+4 \checkmark\\&=7(-1)i-2-3i+8+4 \checkmark\\&=-7i-2-3i+8+4\\&=-10i+10 \checkmark\end{aligned}$$

(3)

1.1.4

$$\begin{aligned}&\frac{\sqrt{-16}-5}{3+5i}\\&=\frac{4i-5}{3+5i}\\&=\frac{4i-5}{3+5i}\times\frac{3-5i}{3-5i} \checkmark\\&=\frac{12i-20i^2-15+25i}{9-15i+15i-25i^2} \checkmark\\&=\frac{12i-20(-1)-15+25i}{9-25(-1)} \checkmark\\&=\frac{12i+20-15+25i}{34} \checkmark\\&=\frac{37i+5}{34} \checkmark\\&=\frac{37}{34}i+\frac{5}{34}\\&=1,088i+0,147\end{aligned}$$

(5)

1.2 1.2.1 $\frac{\sqrt{3}}{2}-\frac{1}{2}i$

$\tan\theta=\frac{\frac{-1}{2}}{\frac{\sqrt{3}}{2}}$ ✓

$\tan\theta=\frac{-\sqrt{3}}{3}$

$\theta=\tan^{-1}\left(\frac{\sqrt{3}}{3}\right)$ ✓

$\theta=30^{\circ}$ (reference angle)

$\therefore\theta=360^{\circ}-30^{\circ}=330^{\circ}$ ✓

$r^2=\left(\frac{\sqrt{3}}{2}\right)^2+\left(\frac{-1}{2}\right)^2$

$r=\sqrt{\left(\frac{\sqrt{3}}{2}\right)^2+\left(\frac{-1}{2}\right)^2}$ ✓

$\therefore r=1$

$rcis\,\theta=1\,cis\,330^{\circ}$ (4)

1.2.2 $2\,cis\,20,2^{\circ}\times30\,cis\,60,8^{\circ}$

$=2\times30\,cis\left(20,2^{\circ}+60,8^{\circ}\right)$ ✓✓

$=60\,cis\,81^{\circ}$ ✓ (3)

1.3 $3\,cis\,133^{\circ}+2\,cis\,20^{\circ}$

$=3i\sin133^{\circ}+3\cos133^{\circ}+2i\sin20^{\circ}+2\cos20^{\circ}$ ✓✓

$=2,194i-2,046+0,684i+1,879$ ✓✓

$=2,878i-0,167$ ✓ (5)

1.4 1.4.1 $\overline{z}=7i-1$ ✓ (1)

1.4.2 $\overline{z}=7i-1$

$\tan\theta=\frac{7}{-1}=-7$

$\theta=\tan^{-1}(7)$

$\theta=81,869^{\circ}$ (ref angle) ✓

$\therefore\theta=180^{\circ}-81,869^{\circ}=98,130^{\circ}$ ✓ (2)

[30]

QUESTION 2

2.1 2.1.1

$$y=2x^3(2x+7)$$
$$=4x^4+14x^3 \checkmark$$
$$\frac{dy}{dx}=16x^3+42x^2 \checkmark\checkmark$$

2.1.2

$$y=\frac{\sqrt{x}+x-\sqrt[3]{x}}{x^3}$$
$$=\frac{\sqrt{x}}{x^3}+\frac{x}{x^3}-\frac{\sqrt[3]{x}}{x^3}$$
$$=x^{\frac{1}{2}-3}+x^{1-3}-x^{\frac{1}{3}-3}$$
$$=x^{\frac{-5}{2}}+x^{-2}-x^{\frac{-8}{3}} \checkmark$$
$$\frac{dy}{dx}=\frac{-5}{2}x^{\frac{-7}{2}}-2x^{-3}+\frac{8}{3}x^{\frac{-11}{3}} \checkmark$$
$$=\frac{-5}{2\sqrt{x^7}}-\frac{2}{x^3}+\frac{8}{3\sqrt[3]{x^{11}}} \checkmark$$

2.1.3

$$y=\frac{x^3}{3}-\sqrt{x}-8x$$
$$=\frac{1}{3}x^3-x^{\frac{1}{2}}-8x \checkmark$$
$$\frac{dy}{dx}=\frac{1}{3}\times 3x^2-\frac{1}{2}x^{\frac{-1}{2}}-8 \checkmark$$
$$=x^2-\frac{1}{2\sqrt{x}}-8 \checkmark$$

(3 x 3) (9)

2.2

$$\lim_{x\to -1}\frac{x^2+x}{x+1}$$
$$=\lim_{x\to -1}\frac{x(x+1)}{x+1}$$
$$=\lim_{x\to -1}x$$
$$=-1 \checkmark\checkmark$$

(2)

2.3 $f(x)=2-x^2$

$f(x+h)=2-(x+h)^2$

$=2-x^2-2xh-h^2$

$f(x+h)-f(x)=2-x^2-2xh-h^2-(2-x^2)$ ✓

$=2-x^2-2xh-h^2-2+x^2$

$=-2xh-h^2$ ✓

$\frac{f(x+h)-f(x)}{h}=\frac{-2xh-h^2}{h}=-2x-h$ ✓

$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=-2x$ ✓ (4)

2.4 $l=1-2h$

$b=0,75-2h$

$=\frac{3}{4}-2h$

$=\frac{1}{4}(3-8h)$

$\therefore v=(1-2h)\times\frac{1}{4}\times(3-8h)\times h$

$=\frac{h}{4}\times(16h^2-14h+3)$

$=\frac{1}{4}\times(16h^3-14h^2+3h)$ ✓

At the maximum or minimum $\frac{dv}{dh}=0$

$\frac{dv}{dh}=\frac{1}{4}(16\times 3h^2-14\times 2h+3)$ ✓

$0=\frac{1}{4}(48h^2-28h+3)$

$0=48h^2-28h+3$

$h=\frac{28\pm\sqrt{(-28)^2-4\times 48\times 3}}{2\times 48}$ ✓

$h=0,141$ or $h=0,441$

$l=1-2h$

$=1-2(0,141)$

$=0,718$ ✓

$b=0,75-2(0,141)$

$=0,468$ ✓ (5)

[20]

QUESTION 3

3.1 3.1.1

$$\frac{x^2-1}{x^3-2x^2-x+2}\div\frac{3x+6}{x^2-4}$$

$$=\frac{x^2-1}{x^3-2x^2-x+2}\times\frac{x^2-4}{3x+6}$$ ✓

$$=\frac{(x-1)(x+1)}{(x^3-2x^2)-(x-2)}\times\frac{x^2-4}{3x+6}$$

$$=\frac{(x-1)(x+1)}{x^2(x-2)-(x-2)-(x-2)}\times\frac{(x-2)(x+2)}{3(x+2)}$$ ✓

$$=\frac{(x^2-1)}{(x^2-1)(x-2)}\times\frac{(x-2)(x+2)}{3(x+2)}$$ ✓

$$=\frac{1}{3}$$ ✓ (4)

3.1.2

$$\frac{x}{x+y}+\frac{y}{x}$$

$$=\frac{x^2+y(x+y)}{(x+y)x}$$ ✓✓

$$=\frac{x^2+xy+y^2}{(x+y)x}$$ ✓ (3)

3.2

$$3x^2-8+2x=-x^2+4$$
$$3x^2+x^2-8+2x-4=0$$ ✓
$$4x^2+2x-12=0$$
$$x=\frac{-2\pm\sqrt{(2)^2-4(4)(-12)}}{2(4)}$$ ✓
$$x=\frac{3}{2}$$ ✓ or $x=-2$ ✓ (4)

3.3 Solve for x:

$x^2-3x-13<-9$

$x^2-3x-13+9<0$ ✓

$x^2-3x-4<0$

$(x-4)(x+1)<0$ ✓

$\therefore x>-1$ ✓ or $x<4$ ✓ (4)

3.4 Solve for x and y:

$y=x^2-3x$

$y-x=-4$

$y=x-4$

$x^2-3x=x-4$ ✓

$x^2-3x-x+4=0$

$x^2-4x+4=0$ ✓

$(x-2)(x-2)=0$ ✓

$x=2$ ✓

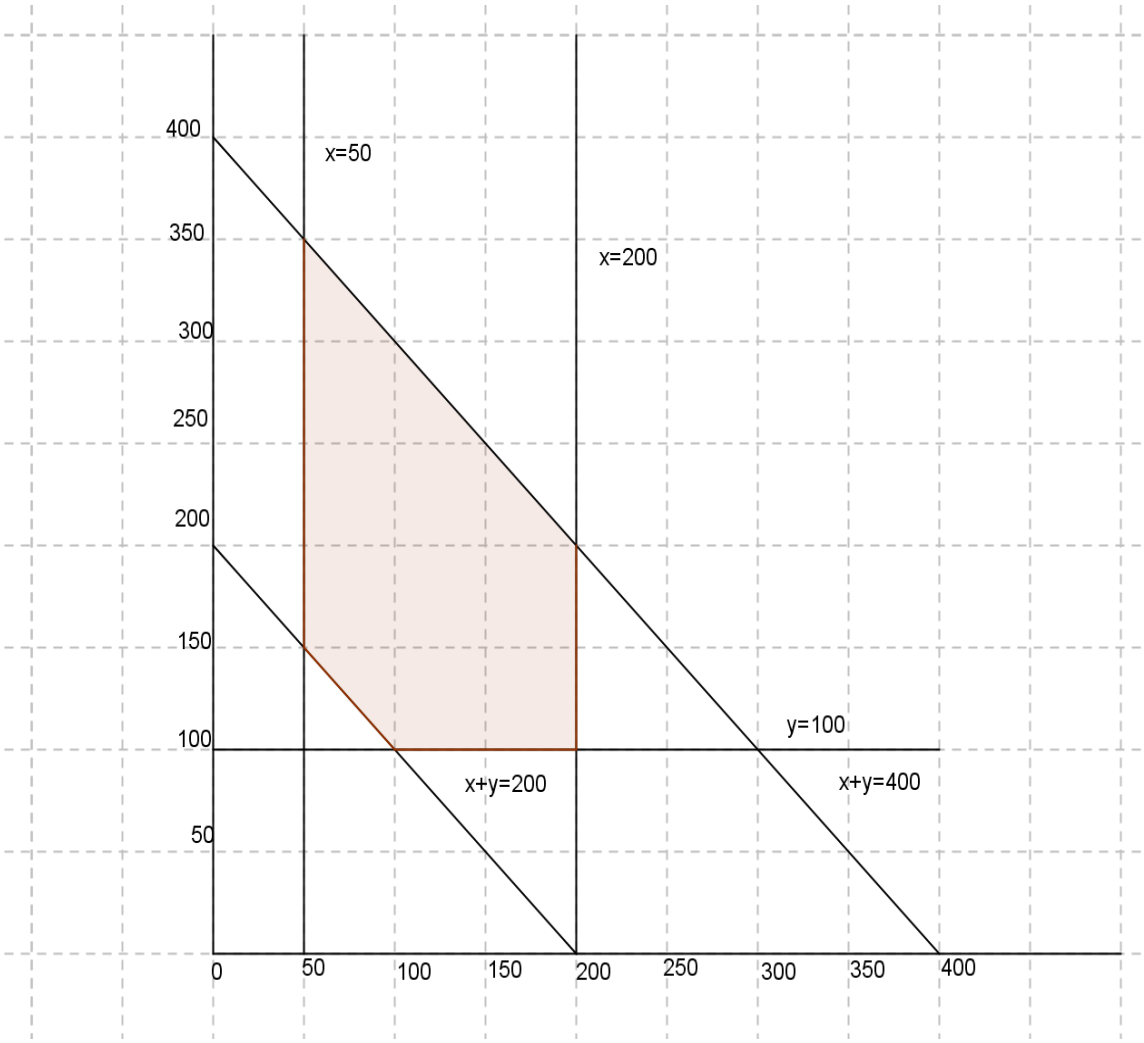
$y=x-4$

$y=2-4$

$y=-2$ ✓ (5)

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3.5 3.5.1 **ADDENDUM A**



(5)

3.5.2 Indicated on the graph✓ (1)

3.5.3 $P = 30x + 10y + 8\ 000$
$(50; 350) = 13\ 000$
$(50; 150) = 11000$ ✓
$(100; 100) = 12\ 000$
$(200; 100) = 15\ 000$

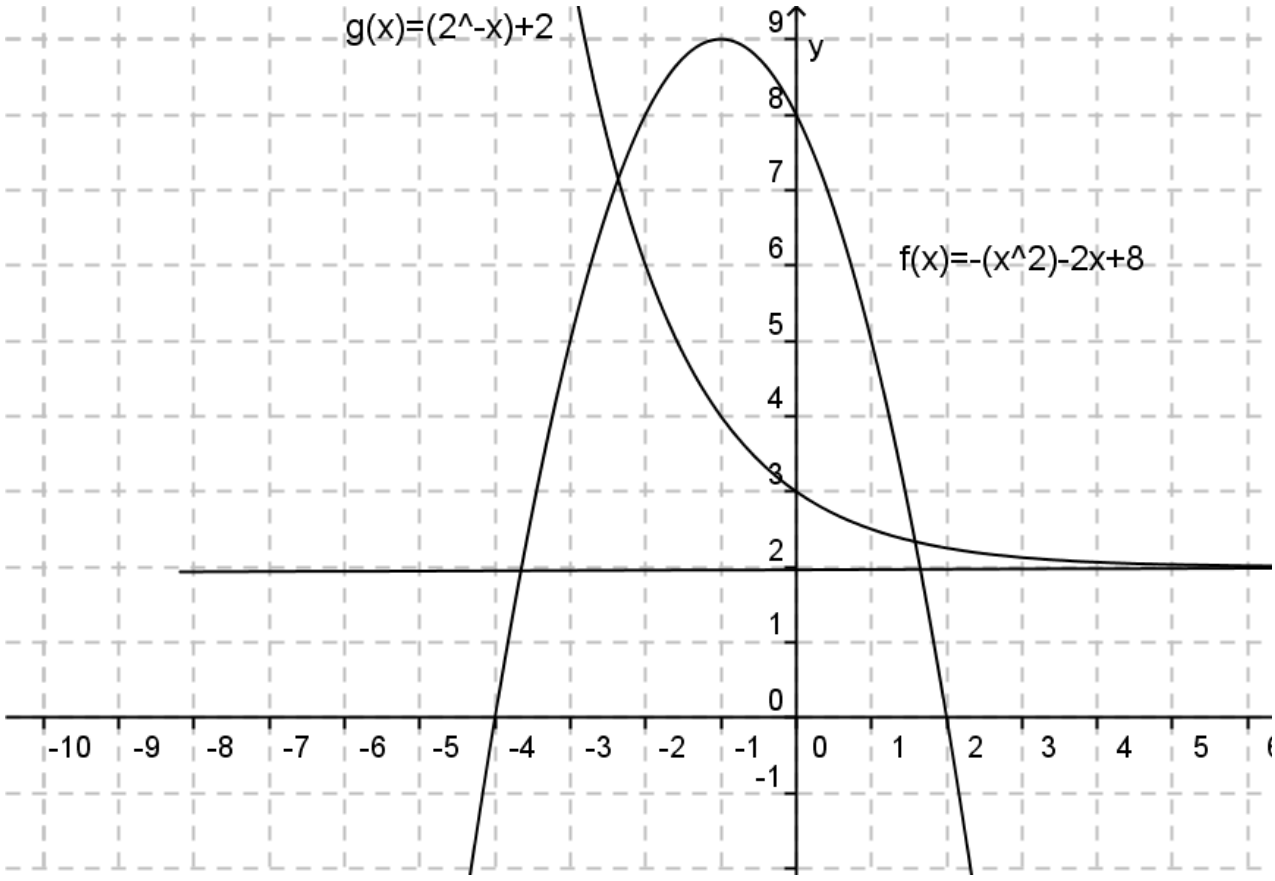
$(200\,;200) = 16\,000$ ✓

Maximum profit is at x = 200✓, y = 200✓ (4)

[30]

QUESTION 4

4.1 4.1.1 **ADDENDUM B**



(8)

4.1.2 y = 2✓ (1)

4.1.3 $x \in \Re$ ✓ (1)

4.2 4.2.1 $x=0$ ✓ (1)

4.2.2 y-intercept: $(0;3)$ ✓
x-intercept: $(1\frac{1}{2};0)$ ✓ (2)

4.2.3 $y=-2;\ x=-1$ ✓✓ (2)

4.2.4 $f(x)=\frac{a}{x+1}-2$
$(0;3):\ 3=\frac{a}{0+1}-2$ ✓
$5=a$ ✓
$f(x)=\frac{5}{x+1}-2$ ✓✓ (4)

4.2.5 $y=2$ ✓ (1)
[20]

TOTAL: 100

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